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COMMERCIAL ITEM DESCRIPTION

SCRUBBING MACHINE, FLOOR, ELECTRIC (BATTERY POWERED)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE

1.1 Scope. This Commercial Item Description covers cabinet type pedestrian electric battery-powered floor scrubbing machines intended for heavy duty floor scrubbing and cleaning. These scrubbing machines shall be designed to recharge on a single phase power source of 120 volts ac, 60 Hz.

2. CLASSIFICATION

2.1 Classification. The scrubbing machines shall be of the following sizes as specified (see 7.2). All size designations are minimums.

Size 1 - 26-inch (660 mm) scrubbing width.
Size 2 - 30-inch (762 mm) scrubbing width.

3. SALIENT CHARACTERISTICS

3.1 Standard Commercial Product. The scrubbing machine shall be the manufacturer's standard commercial product except for any changes necessary for compliance with this CID.

3.2 UL Conformance. The scrubbing machine shall meet the applicable requirements of UL 583, Type E Industrial Truck, and UL 1564. Evidence of conformance shall be a UL label(s) or listing mark(s), or a certified test report(s) (based on tests conducted within the past two years) from an independent testing laboratory acceptable to the Government, indicating the scrubbing machine has passed all applicable tests.

3.3 Construction.

3.3.1 General. Each scrubber shall be complete, lubricated, and ready to operate (minor assembly is permitted). The design shall be simple in construction, shall permit easy accessibility to component parts for cleaning, inspection, adjustment, or replacement, and shall provide reliable operation. Similar component parts and assemblies of identical model number scrubbing machines shall be fully interchangeable. Interchangeability shall not require alteration or modification.

3.3.2 Materials. The materials used for the construction of the scrubbing machine shall be suitable for the purpose intended. Components that are subject to wetting shall be made of corrosion resisting material.

3.3.3 Design. The machine shall be a fully enclosed, cabinet style, key operated, battery-powered, self-propelled pedestrian type. Each machine shall be new and of the manufacturer's latest design. The cabinet shall be provided with resilient, nonmarking replaceable bumpers to prevent the cabinet from marring walls and baseboards. The machine shall be of rugged construction and capable of

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FSC 7910

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withstanding heavy duty commercial use. The machine shall be constructed so that it can be readily disassembled for inspection, adjustment, and replacement of parts.

3.4 Motors. Electric motors shall be of the heavy duty industrial type designed for operation on not less than 24 volts. The motors shall be continuous duty, either semi-enclosed and protected or totally enclosed. Hydraulic motors, if employed, shall be heavy duty and of good commercial quality. The main drive motor shall propel the machine and may drive the scrub brush(es) or a separate brush drive motor may be employed to drive the scrub brush(es).

3.5 Wheels. The machine shall be mounted on two wheels and one or two casters. The machine shall be stable. All wheels shall be bearing equipped and equipped with nonmarking rubber or synthetic tires. The treads of all wheels shall be grooved to eliminate tracking.

3.6 Machine propulsion. All scrubbing machines shall be self-propelled. The drive mechanism shall transmit the full torque of the drive motor without slip. Speed shall be regulated by the operator through a "deadman" control located conveniently at the operator's station. The machine shall be capable of powered reverse without requiring the operator to leave the control position.

3.7 Controls. All controls necessary for machine operation shall be conveniently located at the operator's station.

3.7.1 Mechanical controls. All mechanical controls shall be smooth in operation and shall not require excessive force for actuation. Mechanical control positions shall be detented to prevent accidental dislodgment.

3.7.2 Locking switch. The scrubber shall have a key-operated locking switch to prevent unauthorized use.

3.7.3 Indicators. Each machine shall be equipped with a battery condition indicator. The indicator shall be located at the operator's station within the operator's normal field of vision.

3.7.4 Hydraulics. All components in the hydraulic system, if employed, shall be heavy duty and of good commercial quality. Hydraulic pistons and similar wiped components shall be resistant to the corrosive effects of the cleaning solutions and solvents employed in floor maintenance operations.

3.8 Machine sizes. Machine dimensions and weight shall be in accordance with table I for the sizes indicated.

Table I. Scrubbing machine sizes and performance

Size	Scrubbing width range (inches)	Machine dimensions			Tank Capacities		Machine weight less battery (max.) (lb)	Brush load (min.) (lb)	Scrubbing performance (min.) sq-ft/hr
		Length (max.) (inches)	Width (max.) (inches)	Height (max.) (inches)	Recovery (min.) (gallons)	Solution (min.) (gallons)			
1	26-29 (660-737 mm)	60 (1.52 m)	32 (813 mm)	46 (1.17 m)	15 (56.8 L)	15 (56.8 L)	600 (272 kg)	115 (52.1 kg)	16,000 (1,490 m^2/hr)
2	30-35 (762-889 mm)	65 (1.65 m)	36 (914 mm)	46 (1.17 m)	20 (75.7 L)	22 (83.3 L)	700 (317 kg)	150 (68.0 kg)	20,000 (1,860 m^2/hr)

3.9 Machine brush requirements.

3.9.1 Brushes and floor pads. The machine shall be furnished with the type and number of scrubbing brushes necessary for operation. Scrubbing brushes shall be readily replaceable without the use of special tools. The machine shall be

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compatible for use with floor pads having 3-1/2 ± 1/8 inch (88.9 ± 3.18 mm) diameter center holes.

3.9.2 Brush housing. The brush housing shall afford adequate protection from the rotating brush(es). The housing (if protruding) shall be provided with resilient, nonmarking replaceable bumpers. Curtains or baffles to prevent slinging the cleaning solution beyond the periphery of the housing shall be provided.

3.9.3 Brush drive. The power transmission system shall be capable of delivering the necessary torque to maintain a brush speed of no less than 100 RPM at the maximum brush load selectable by the operator without slip.

3.9.4 Brush mounting. Twin disc machines shall have brush overlap to eliminate gaps in the cleaning swath. The brush holders shall secure the brushes but shall permit brush changing with minimum effort and without special tools. The holders shall provide for brush positioning and loading. Positioning and loading shall be controllable by the operator. When in contact with the floor, minimum brush loading shall be as specified in table I. The brush holder shall be self-aligning or articulated to permit full brush contact on uneven floors.

3.10 Squeegee. A nonmarking squeegee shall be located aft of the brushes. The squeegee material shall be resilient, nonhardening, oil-resistant and impervious to common cleaning detergents and solvents. The squeegee length shall be one inch (25.4 mm) greater (minimum) than the cleaning swath. The squeegee shall be spring loaded or equipped with rollers to permit operation next to walls or protrusions. The squeegee shall be of sufficient height to prevent solution being thrown over the blade by the brush(es). The squeegee assembly shall be spring loaded to permit surface contact over uneven floors and shall be rigged to follow the brush swath. The assembly shall be capable of being raised or lowered by the operator from his station through actuation of convenient hand or foot controls.

3.11 Solution and recovery tanks. The tank capacities shall be as specified in table I. Tank materials, coatings, coverings, etc., shall be immune to attack by common detergents and solvents utilized in floor maintenance. Where weldments are used in tank construction, electrode or filler rod material shall approximate the base metal. Covers or baffles shall be provided to prevent solution from splashing out of the tanks. The tanks shall be easily accessible for cleaning and removal.

3.11.1 Clean solution tank. The clean solution tank shall be mounted in the machine so that it can be easily and rapidly filled. The flow of solution from the tank to the brush shall be controlled by means of a regulating type valve. The valve shall be controlled from the operator's position. The solution from the tank shall be fed through pipes or hoses to the center of each brush. The feed lines shall be equipped with an accessible strainer. The tank shall be provided with an auxiliary valve or similar means or method for quick emptying.

3.11.2 Recovery tank. The recovery tank shall be provided with an automatic cutoff mechanism which will deactivate the vacuum producer system when the recovered solution reaches a predetermined level. Some means for sludge collection shall be provided in this tank; if a dump valve is employed, the valve shall permit quick emptying and shall be located no closer than 6 inches (152 mm) from the floor when the tank is installed in the machine. The sludge trap or tank shall be easily removable for cleaning. A dump valve is not required if alternate methods for tank cleaning and emptying are shown to be acceptable.

3.12 Operating handle. The machine shall be equipped with an operator's control handle. The handle shall be located at the rear of the machine and in a position convenient to the operator when walking and guiding the machine. The handle shall be provided with suitable rubber or plastic hand grips.

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3.13 Battery compartment. The battery compartment shall be adequately ventilated and of ample size to house the batteries required and so constructed as to prevent movement of and damage to the batteries during scrubbing and transporting operations. The compartment cover shall be hinged or readily removable and when opened shall expose the entire battery. The inside surface of the cover, if of metal, shall be lined or coated with a suitable insulating material. In addition to the hinged or removable cover, one or both sides of the battery compartment may be hinged or removable at the option of the contractor. The compartment shall be constructed so that the batteries can be readily inspected, serviced, removed, and installed.

3.14 Battery. The batteries shall be furnished by the supplier of the machines. The nominal potential of the batteries shall not be less than 24 volts. Capacities required will be determined by the contractor to satisfy the performance requirements of table I but in no case shall they be less than 200 ampere-hours.

3.15 Battery charger. A battery charger shall be furnished with each machine. The charger may be furnished as an accessory or may be built in. The charger shall conform to the requirements of UL 1564. The charger shall have a capacity of not less than 15 amperes. The charger shall be provided with a timer that can be set to charge the batteries automatically.

3.16 Maintenance. Periodic lubrication and maintenance requirements shall be kept to a minimum. Provisions for lubrication shall be incorporated where required. Lubricated parts shall be adequately sealed or shielded against entry of foreign matter. Lubricant retainers shall be employed where leakage may be anticipated. The machine shall be completely lubricated prior to delivery. Maintenance shall not require special tooling. All components and assemblies subject to wear or replacement shall be readily accessible.

3.16.1 Hydraulic system requirements. Periodic maintenance requirements on the hydraulic system shall be kept to a minimum. A filter shall be required between the reservoir and the pump inlet. Filtration to 10 microns is required. The filter shall be equipped with a bypass indicator. Servicing shall be limited to replacement of the filter element and fluid checks.

3.17 Finish. The scrubbing machines shall have all ferrous components finished with a durable enamel, lacquer, or other suitable form of corrosion protection. Stainless steels, glass fiber, rubber, or plastic parts do not require finishing. Colors and finishes normally applied for commercial customers will be acceptable.

3.18 Marking. Each scrubbing machine shall be marked in a permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.19 Instructions. The manufacturer shall furnish with each scrubbing machine a booklet or pamphlet giving complete instructions for the operation, lubrication, adjustment, maintenance, and care of the machine. An illustrated parts breakdown and a repair parts list shall also be furnished with each machine.

3.20 Workmanship. The scrubber and all accessories shall be constructed, assembled, and finished in a manner to assure good quality equipment of a neat general appearance and free from imperfections which may affect durability, serviceability, and appearance.

3.21 Measurement system. The values stated in inch-pound units are to be regarded as the standard. The metric values stated in parentheses are provided to an accuracy of three significant digits and are for informational purposes only.

3.22 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units,

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providing they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

3.23 Commercial item. The use of the term "commercial item" in this document does not imply that any item or items offered are not required to conform with all requirements specified herein.

3.24 Performance. Tests for scrubbing, machine maneuverability and speed, and mechanical operation shall be conducted on a swept floor surface consisting of smooth concrete. Scrubbing effectiveness shall be relative to type of soilage, and the chemical compounds utilized.

3.24.1 Scrubbing. The machine shall be equipped with suitable brushes and the solution tank filled with a proper cleaning solution.The machine shall exhibit good performance and simplicity of operation. The brush(es) shall scrub evenly over the required scrubbing width. The squeegee shall collect all solution deposited by the brush(es) in the squeegee swath whether in a straight line or going through 90° turns. The vacuum producer shall recover all the solution collected by the squeegee. Scrubbing performance shall be in accordance with table I. Scrubbing performance shall be on the basis of a one-hour test run to determine conformance to the scrubbing area requirements specified in table I. In lieu of running the test for one hour, the test may be run for not less than 20 minutes and the results extrapolated to the one hour basis. The floor passed over and immediately to the rear of the machine shall be observed to determine that it is only damp and that no water or puddles remain on the floor. Observation shall be made to determine that the flow of cleaning solution from the clean solution tank to the brush is uniform and that the flow can be properly regulated by the operator. The clean solution tank, the recovery tank, feed lines, valves, and connections shall be examined for leakage.

3.24.2 Machine maneuverability and speed. The machine shall be negotiated through random turns at various speeds and accelerations to determine that it is easily maneuverable. The self-propulsion speed of the machine shall be operator-regulated and continuously variable from 0 to not less than 160 feet per minute (fpm), (48.8 meters per minute) on smooth, level, concrete floors with the brushes scrubbing at the minimum load pressure specified in table I. Maximum speed under these conditions shall be 264 fpm (80.5 meters per minute). The machine shall be capable of negotiating an 8 percent grade at no less than 88 fpm (26.8 meters per minute) while scrubbing at the specified minimum load pressures.

3.24.3 Mechanical operation. Each electrical and mechanical control shall be actuated to determine proper operation. The drive motor control shall be actuated to determine that the propulsion system accelerates and decelerates the machine smoothly and regularly through the entire speed range. While in motion the drive control shall be released. The machine shall decelerate smoothly to a stop and the machine shall not creep until the control is again actuated.

3.24.4 Vacuum producer. The vacuum producer shall be capable of developing 16 inches (406 mm) of water lift through a 7/8 inch (22.2 mm) diameter orifice for a continuous operation of 4 hours. The vacuum producer shall recover the cleaning solution collected by the squeegee and deposit it in the recovery tank. Vacuum testing shall be in accordance with ASTM Standard F558. Shortening or elimination of ducting for test purposes is not permitted.

3.24.5 Noise. The main drive motor, brushes, and vacuum producer shall be in operation and the propulsion mechanism shall be disabled. Six A weighted sound level readings shall be taken each at a different point at a 6-foot radius from the vacuum producer exhaust port at 60° circumferencial intervals. The average of the readings shall not be greater than 70 dbA.

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4. REGULATORY REQUIREMENTS

4.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Referenced Documents. The documents referenced in this CID shall be the issues in effect on the date of issuance of the invitation for bids or request for proposals unless otherwise specified. The documents form a part of this CID to the extent specified. In the event of a conflict between this CID and the documents referenced herein, this CID shall take precedence.

American Society for Testing and Materials Standards:

ASTM F558 - Standard Test Method for Measuring Air Performance Characteristics of Vacuum Cleaners.

Application for copies of ASTM standards should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

Underwriters Laboratories Inc., Standards:

UL 583 - Safety Electric-Battery-Powered Industrial Trucks.

UL 1564 - Safety Industrial Battery Chargers.

Application for copies of UL standards should be addressed to Underwriters Laboratories Inc., Publication Stock, 333 Pfingsten Road, Northbrook, IL 60062.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents (if applicable).

a. Title, number and date of this commercial item description.
b. Size.
c. The preservation, packing, and marking desired.

MILITARY INTERESTS:

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